A formal Jacobian ideal certificate for a mixed quartic

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Abstract

For the explicit ring $A = \mathbb{Q}[x_0, x_1, x_2, x_3]/(\sum_i x_i^4 + \sum_{i<j} x_i^2 x_j^2)$, we give a complete Lean-checked construction of a differential operator of order at most two outside the full algebra generated by multiplication operators and all derivations. A finite rational coefficient functional detects its failure to preserve the Jacobian ideal. The proof covers arbitrary finite derivation words, including cancellation among words of different lengths. We also give a written extension of the Hessian construction to isolated ordinary homogeneous hypersurfaces, using the classical complete-intersection socle theorem. That general extension is not formalized here. The three-variable operator formula is already in the literature, and publication priority for the higher-dimensional argument has not been established. This note does not resolve the unrestricted Nakai conjecture.

1 Scope and prior work

For a commutative algebra A over a characteristic-zero field k, let

$$\Delta_k(A) = k\langle m_a, \delta : a \in A,\ \delta \in \mathrm{Der}_k(A)\rangle \subseteq \mathrm{End}_k(A), \qquad m_a(b) = ab.$$

This is an algebra with no bound on word length. If $[T, m_a] = Tm_a - m_aT$, the condition

$$[[[T, m_a], m_b], m_c] = 0 \qquad (a, b, c \in A) \tag{1}$$

is the usual faithful endomorphism definition of differential order at most two. Our formal statement produces a T satisfying (1) and $T \notin \Delta_k(A)$.

This distinction matters. Exclusion from the module spanned by derivations and products of two derivations does not by itself exclude membership in $\Delta_k(A)$: longer words could cancel in their highest-order parts. The introduction and principal theorem of the 8 October 2026 version of Yau–Zhu–Zuo [2] explicitly use generation order by order and $\mathrm{der}^2(A) \neq \mathrm{Der}^2(A)$. We do not replace the unrestricted algebra by that shorter-word module.

Prior formula and method. The operator below is not claimed to be a new formula in three variables. Diethorn–Jeffries–Miller–Packauskas–Pollitz–Rahmati–Vassiliadou [1, Sections 4.1.3 and 4.5, equations (33), (35), Theorem 4.5.1] give precisely $Q_r/(d-1)^2$ in that case, after conversion from divided partials and inclusion of the first-order correction. Their introduction also discusses the earlier work of Vigué. Stable-ideal separation is a known method; it occurs, for example, in the Fermat-type work of Brumatti–Veloso [3]. Hessian-adjugate constructions also overlap [2]. Traves [5] proves the full conjecture for varieties with smooth normalization. Consequently neither a general novelty claim nor a claim to have solved a previously unverified singularity class follows from this note. The formal verification and explicit full-word certificate are the precisely documented outputs.

2 The fixed quartic theorem

Put $P = \mathbb{Q}[x_0, x_1, x_2, x_3]$ and

$$F = \sum_{i=0}^{3} x_i^4 + \sum_{0 \le i < j \le 3} x_i^2 x_j^2. \tag{2}$$

Let $G_i = \partial_i F$, $H = (\partial_i \partial_j F)$, $C = \mathrm{adj}(H)$, $h = \det H$, $a_i = C_{i0}$, and

$$B_{ij} = a_i x_j + x_i a_j - x_0 C_{ij}.$$

Define the explicit polynomial operator

$$\mathcal{Q}(p) = \sum_{i,j=0}^{3} B_{ij} \partial_i \partial_j p + 6 \sum_{i=0}^{3} a_i \partial_i p. \tag{3}$$

The partials are ordinary, not divided. Both ordered off-diagonal terms occur in the double sum.

Theorem 2.1 (Fully formalized fixed case)**.** *Let $A = P/(F)$ and let $\pi : P \to A$ be the quotient map. There is $Q \in \mathrm{End}_{\mathbb{Q}}(A)$ such that*

$$Q(\pi(p)) = \pi(\mathcal{Q}(p)) \quad (p \in P), \qquad [[[Q, m_u], m_v], m_w] = 0 \quad (u, v, w \in A), \qquad Q \notin \Delta_{\mathbb{Q}}(A).$$

The Lean declarations are

```
NakaiQuartic.quartic_full_operator_separation
NakaiExpected.expected_full_operator_separation
```

The latter restates the literal ten-term polynomial, multiplication maps, commutators, explicit polynomial operator, and unrestricted generating set independently of the implementation abbreviations. Neither statement assumes the desired nonmembership conclusion. Sections 3–5 provide the mathematical proof and its certificates.

3 The ideal preserved by every derivation

Lemma 3.1. *Let $P = k[x_1, \ldots, x_n]$, $f \in P$, and $A = P/(f)$. The image J_A of the gradient ideal $J = (f_1, \ldots, f_n)$ is preserved by every derivation of A. Every element of $\Delta_k(A)$ therefore preserves J_A.*

Proof. For $\delta \in \mathrm{Der}_k(A)$, choose representatives $u_i \in P$ of $\delta(\overline{x_i})$, and put $\widetilde{\delta} = \sum_i u_i \partial_i$. The two induced derivations agree on all polynomial generators, hence agree on P. In particular $\widetilde{\delta}(f) = vf$ for some $v \in P$. Differentiating this equality gives

$$\widetilde{\delta}(f_j) = v f_j + (\partial_j v) f - \sum_i (\partial_j u_i) f_i. \tag{4}$$

Thus $\widetilde{\delta}$ preserves $(f) + J$, and δ preserves J_A. Multiplication maps also preserve J_A. The endomorphisms preserving a fixed ideal form a k-subalgebra, so they contain the entire algebra generated by these maps. This covers every finite word and every linear combination of words. □

The formal framework uses the additional hypothesis $f \in J$, which is proved explicitly for F by Euler's identity. It formalizes the polynomial coefficient lift of every quotient derivation; no generation theorem for Koszul derivations is assumed.

4 A written homogeneous extension

This section is a written proof, not a claim that the full class has been formalized. Let k have characteristic zero, let $n \geq 3$, and let $f \in P = k[x_1, \ldots, x_n]$ be homogeneous of ordinary degree $d \geq 2$. Write $X = (x_i)^t$, $H = \mathrm{Hess}(f)$, $C = \mathrm{adj}(H)$, and $h = \det H$. Fix r, put $a = Ce_r$, and set

$$c = (n-2)(d-1), \qquad B = aX^t + Xa^t - x_r C, \qquad Q_r(p) = \sum_{i,j} B_{ij}\partial_i\partial_j p + c\sum_i C_{ir}\partial_i p. \tag{5}$$

Proposition 4.1 (Exact polynomial identities)**.** *For every $p \in P$ and every j,*

$$Q_r(fp) = f\left(Q_r(p) + 2d\sum_i C_{ir}\partial_i p\right), \tag{6}$$

$$Q_r(f_j) = (nd - n - d)\delta_{rj}h - (d-1)\sum_i (\partial_j C_{ir})f_i. \tag{7}$$

Proof. Euler's identity and its derivative give

$$X^t\nabla f = df, \qquad HX = (d-1)\nabla f, \qquad C\nabla f = \frac{hX}{d-1}.$$

Since H and C are symmetric,

$$B\nabla f = dfa, \qquad \mathrm{tr}(BH) = 2a^tHX - x_r\,\mathrm{tr}(CH) = (2-n)x_r h.$$

It follows that $Q_r(f) = 0$. The product rule and symmetry of B now yield

$$Q_r(fp) = fQ_r(p) + pQ_r(f) + 2(B\nabla f)^t\nabla p,$$

which proves (6) on every polynomial, not just on generators.

For (7), the entries of H have degree $d-2$, so $\sum_\ell x_\ell\partial_\ell H_{ij} = (d-2)H_{ij}$. The two rank-one terms in B, followed by the first-order part, contribute $[2(d-2)+c]\delta_{rj}h$ to $Q_r(f_j)$. Jacobi's differentiation formula for the determinant gives the remaining term $-x_r\partial_j h$. On the other hand, differentiating $(d-1)\sum_i C_{ir}f_i = x_r h$ gives

$$x_r\partial_j h = (d-2)\delta_{rj}h + (d-1)\sum_i (\partial_j C_{ir})f_i.$$

Substitution proves the formula, since $d - 2 + c = nd - n - d$. □

Theorem 4.2 (Written homogeneous statement)**.** *Assume additionally that $P/J(f)$ is finite-dimensional. Each operator Q_r in* (5) *descends to $A = P/(f)$, has order at most two, and lies outside $\Delta_k(A)$.*

Proof. Equation (6) proves ideal preservation and quotient descent. The displayed operator has differential order at most two; the triple commutator identity passes to the quotient by surjectivity.

Euler's identity puts f in $J(f)$. The gradient generators have positive degree and define a zero-dimensional ideal, so they form a regular sequence and $P/J(f)$ is a nonzero Artinian local complete intersection supported at the origin. It is already complete and has presentation $k[[x_1, \ldots, x_n]]/(f_1, \ldots, f_n)$. The classical complete-intersection Jacobian–socle theorem, in the form of Eisenbud–Ulrich [4, Theorem 7.1, p. 210], says that its Kähler different is nonzero and equals its socle. Here the different is generated by $\det(\partial_j f_i) = h$. Thus $h \notin J(f)$.

The integer $nd - n - d = (n-1)(d-1) - 1$ is positive under the stated hypotheses. Equation (7) shows that Q_r sends $\overline{f_r} \in J_A$ outside J_A. Lemma 3.1 therefore excludes Q_r from the full generated algebra. □

Remark 4.3. The isolatedness hypothesis is used for the classical socle conclusion, not for the exact operator identities or derivation stability. Ordinary homogeneity is essential to the argument given here; no weighted-homogeneous extension is inferred by changing terminology.

5 A finite certificate over the rationals

For the fixed quartic (2), put $S = \sum_i x_i^2$. Direct calculation gives

$$G_i = 2x_i(S + x_i^2), \qquad H_{ii} = 2S + 10x_i^2, \qquad H_{ij} = 4x_i x_j \ (i \neq j).$$

The specialized identities, checked in Lean for the actual Hessian adjugate, are

$$\mathcal{Q}(Fp) = F\left(\mathcal{Q}(p) + 8\sum_i a_i \partial_i p\right), \tag{8}$$

$$\mathcal{Q}(G_0) = 8h - 3\sum_i (\partial_0 a_i) G_i. \tag{9}$$

Define a $\mathbb{Q}$-linear functional $\lambda : P \to \mathbb{Q}$ by its values on monomials. It vanishes except in degree eight on the following sorted exponent types:

Exponent type	$\lambda(x^\alpha)$	coefficient of h	multiplicity
$(8,0,0,0)$	3	96	4
$(6,2,0,0)$	-2	720	12
$(4,4,0,0)$	-2	1248	6
$(4,2,2,0)$	3	4128	12
$(2,2,2,2)$	-12	16272	1

Lemma 5.1. *For all i and all $p \in P$, $\lambda(G_i p) = 0$, whereas $\lambda(h) = -77760$.*

Proof. By linearity and degree, the annihilation calculation reduces to degree-five monomials p. Only those odd in x_i and even in the other variables can contribute. Their types are x_i^5, $x_i^3 x_j^2$, $x_i x_j^4$, and $x_i x_j^2 x_k^2$, with distinct indicated indices. The corresponding evaluations are

$$4(3) + 6(-2) = 0, \quad 4(-2) + 2(-2) + 4(3) = 0, \quad 4(-2) + 2(-2) + 4(3) = 0,$$

$$4(3) + 2(3) + 2(3) + 2(-12) = 0.$$

All other parity patterns vanish termwise. Expanding the determinant gives the coefficients in the table; consequently

$$\lambda(h) = 4 \cdot 3 \cdot 96 - 12 \cdot 2 \cdot 720 - 6 \cdot 2 \cdot 1248 + 12 \cdot 3 \cdot 4128 - 12 \cdot 16272 = -77760.$$

The formal proof instead uses the coefficient formula for a monomial times an arbitrary polynomial, proving $\lambda(G_i p) = 0$ directly for every p. The full 35-monomial determinant expansion and all functional values are proved by kernel-checked algebraic normalization. □

Proof of Theorem 2.1. Equation (8) gives an actual linear operator on the quotient. Its triple commutators vanish because $\mathcal{Q}$ is a sum of polynomial coefficients times partial-derivative compositions of length at most two. The quotient map transports this identity. Euler's identity gives $F \in J$, and Lemma 5.1 gives $h \notin J$. More explicitly, (9) yields

$$\lambda(\mathcal{Q}(G_0)) = 8\lambda(h) = -622080 \neq 0,$$

so $\mathcal{Q}(G_0) \notin J$. As $(F) \subseteq J$, this remains an escape from the image of J in the actual quotient. Apply Lemma 3.1. The concrete proof requires no socle theorem. □

6 Geometric checks outside the formal theorem

These observations are written proofs and are not asserted as additional Lean declarations. They explain the example's geometric setting. For $F_n = \sum_i x_i^4 + \sum_{i<j} x_i^2 x_j^2$, $n \geq 3$, a common zero of the gradient over an algebraic closure satisfies $x_i(S + x_i^2) = 0$. If its nonzero support has size r, then $x_i^2 = -S$ on that support, whence $S = -rS$. Characteristic zero gives $S = 0$, and every coordinate vanishes. Thus the critical point is isolated.

There is no nontrivial linear Thom–Sebastiani splitting. Such a splitting would produce a nontrivial constant idempotent matrix M with $H(X)M = M^t H(X)$. Comparing coefficients of x_k^2 gives

$$(2 + 10\delta_{ik})M_{ij} = (2 + 10\delta_{jk})M_{ji}.$$

For $i \neq j$, choose first k distinct from both and then $k = i$. This forces $M_{ij} = M_{ji} = 0$. Comparing $x_i x_j$ then equates all diagonal entries. Thus M is scalar and has no nontrivial idempotent. This excludes linear separated-variable forms; it is not a complete analytic/contact-equivalence classification or a priority argument.

7 Formal specification and reproducibility

The formalization uses a four-variable multivariate polynomial algebra over $\mathbb{Q}$, the actual ideal quotient, `Derivation`, and `Algebra.adjoin`. The generated algebra includes every multiplication map and every derivation of that quotient; no hand-picked list of derivations substitutes for it. The generic assembly theorem proves that four polynomial certificates imply full quotient separation: $f \in J(f)$, preservation of every multiple of f, the triple-commutator identity on the polynomial ring, and an actual gradient-generator escape. All four are discharged for (2).

The checked source closure has fifteen modules. The final declarations report only the standard axioms `propext`, `Classical.choice`, and `Quot.sound`; there are no custom axioms, unfinished proof placeholders, or native computation oracles. The pinned versions are:

```
Lean 4.32.2: f3b06c705e6c85f5314019d5d3baab0fec5b580c
mathlib: 905b95818eb32af7874a58b427f50c1711a5e96c
```

The accompanying manifest records source hashes, the dependency order, successful checks, and the independent replay. A clean rebuild of all fifteen modules and the expanded final theorem also succeeded, using matching copied source snapshots and a fresh output directory without the original object files. Its log is included in the package. The supplied build script uses one Lean thread and fixed time and memory guards.

Theorem 2.1 is the full formal result. Theorem 4.2 remains a written theorem with the classical import stated explicitly. No unrestricted Nakai result and no mathematical publication priority is claimed.

References

[1] R. N. Diethorn, J. Jeffries, C. Miller, N. Packauskas, J. Pollitz, H. Rahmati, and S. Vassiliadou, *Resolutions of differential operators of low order for an isolated hypersurface singularity*, arXiv:2209.13110v3, 16 November 2023. `https://arxiv.org/html/2209.13110v3`.

[2] S. S.-T. Yau, Q. Zhu, and H. Zuo, *Nakai conjecture for isolated weighted homogeneous hypersurface singularities*, arXiv:2604.24508v4, 8 October 2026. `https://arxiv.org/html/2604.24508v4`.

[3] P. R. Brumatti and M. O. Veloso, *A note on Nakai's conjecture for the ring* $K[X_1, \ldots, X_n]/(a_1X_1^m + \cdots + a_nX_n^m)$, Colloquium Mathematicum **123** (2011), 277–283. https://doi.org/10.4064/cm123-2-10.

[4] D. Eisenbud and B. Ulrich, *Duality and socle generators for residual intersections*, Journal für die reine und angewandte Mathematik **756** (2019), 183–226, especially Theorem 7.1, p. 210. https://par.nsf.gov/servlets/purl/10282767.

[5] W. N. Traves, *Nakai's conjecture for varieties smoothed by normalization*, Proceedings of the American Mathematical Society **127** (1999), 2245–2248. https://www.usna.edu/Users/math/traves/_files/documents/nakaisconjecture.pdf.